# Emily Hough-Kovacs

+1 (603) 484-2534 | San Francisco, CA | emilyhoughkovacs@gmail.com
www.emilyhk.com | linkedin.com/in/emilyhoughkovacs

## Summary

Senior data scientist with 10+ years turning large-scale product data into rigorous measurement and decisions at Reddit, Meta, and Spotify. Now transitioning full-time into machine learning and AI, from hands-on classical ML (random forest ensembles, NMF, word2vec embeddings) to LLM pipelines on the Anthropic API. Strong Python foundation; building depth in deep learning and transformer-based language models.

## Skills & Tools

**Languages & Libraries:** Python (pandas, NumPy, scikit-learn, statsmodels, matplotlib), SQL
**Data Platforms & Pipelines:** GCP BigQuery, PostgreSQL, Hive, HDFS, Airflow, ETL, data pipelines
**Statistics & Analytics:** statistical modeling, metric design, Tableau
**AI / ML Tooling:** LLM APIs (OpenAI, Anthropic), embeddings & vector search, Claude Code, Cursor
**Engineering:** Git, GitHub, Flask

## Experience

**Reddit** — Senior Data Scientist, Ads Marketplace March 2026 – Present

- Own data science for the identity matching and attribution domain, standing up its measurement foundation (metric taxonomy, pipelines, reporting) in a system that has no true measurement today; validated engineering's opportunity-sizing against source data.
- Built two Airflow ETL pipelines powering an internal dashboard for ads-engineering data amendments and make-goods during account incident remediation, keeping billed and reported figures consistent while cutting manual ticket overhead.
- Built a supply-and-demand monitoring dashboard surfacing contextual ad-to-post relevance signals across the marketplace, informing inventory and pacing decisions.

**Meta** — Senior Data Scientist, Reality Labs July 2024 – August 2025

- Built and evaluated a random forest ensemble predicting user purchase intent, assessing performance tradeoffs.
- Built long-range forecasts for Lifestyle and Entertainment teams' key metrics, informing roadmap and resourcing decisions.
- Ran observational analyses of MR vs. fully immersive VR effectiveness, quantifying the impact of physical space on user engagement.

**Spotify** — Data Scientist Aug 2022 – March 2024
May 2016 – May 2021

- Pioneered metrics suite for the creator side of the music marketplace, adopted as source of truth across creator-facing teams.
- Evaluated and monitored time-to-delivery of podcast content, surfacing supply-chain bottlenecks for podcast partnerships.
- Built and maintained Tableau dashboards tracking performance of internal and external editorial tools.
- Produced creator-side insights for Spotify Wrapped '22 and '23 (top editorial tracks, artists, albums), surfaced in Spotify's public, consumer-facing product.

**Tumblr** — Product Analyst September 2014 – October 2015

- Extracted business insights for a platform serving 250M+ blogs and 60M+ posts/day; automated reporting and built a Flask app to execute and email ad-hoc Hive jobs.

## Education

**Barnard College, Columbia University** — B.A., Mathematics / Computer Science May 2014

## Projects & Writing

**research-watcher** — daily LLM pipeline (Python, Claude API, GitHub Actions) that ranks and summarizes new AI papers.
**Agentic Customer Behavior Simulator** — persona-based agents model real customer purchase intent and behavior.
**Build Your Own LLM** — writeup from a hands-on workshop building a GPT-style model from scratch.
**Semantle** — word-guessing game (Python, gensim, FastAPI): word2vec embeddings + cosine similarity.
**Playlistr** — playlist generator (Python, MongoDB): NMF over Rap Genius song metadata surfaces similar songs.